

Modeling general developable surfaces in CeDG: a preliminary study

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Abstract. Computer extended Descriptive Geometry is a novel approach for the construction of 3D parametric models by their projections, which address the spatial challenges associated with Descriptive Geometry. This paper examines the potential of CeDG to model general developable surfaces, with a focus on those with warped and non-proper (infinite location zones) regression edges. We designed and assessed a novel, discrete preliminary algorithm with two case studies: (i) a developable helicoid and (ii) a convolute with planar and warped directrix curves. The projections and flat pattern of the developable helicoid were exact, whereas the flat pattern of the convolute was approximated by an accurate discrete solution. Our results suggest that CeDG can be expanded to implement a robust final method for modeling developable surfaces and their flattened patterns, supporting dynamic parameterization.

Keywords: Computer extended Descriptive Geometry, Developable surfaces, Flat patterns, Regression edge, CAD

1 Introduction

CeDG technology is designed to enable the construction of computational three-dimensional parametric models that address the spatial analysis challenges inherent in Descriptive Geometry (DG) [1]. Each model created with CeDG consists of a set of interconnected graphic-mathematical objects. These objects are organized in accordance with a particular construction sequence.

According to the results of earlier studies, CeDG enhances the precision and scope of generating the associated flattened patterns of radiated spatial surfaces, specifically for cones and cylinders, surpassing conventional CAD techniques [2, 3]. However, developable ruled surfaces encompass a wider range of technically notable surfaces, including the developable helicoid, convolutes, and surfaces of constant slope. Although these surfaces are well understood in DG, their resolution under classical manual techniques remains highly complex. Obtaining their accurate flattened patterns becomes unfeasible when the regression edge is warped, as evidenced by their missing in classic DG treatises [4] or by their gross approximations by the method of triangulation [5].

Despite the significance of developable surfaces in domains such as naval, textile, and architectural design, as well as in fields like animation and augmented reality, CAD

tools still encounter difficulties in fully managing these surfaces due to their non-linear nature. Most computer-based methods employ the C^2 smoothness property of such surfaces. These methods utilize either discrete triangular approximations or, more recently, spline-based discrete approximations. Spline-based discrete approximation is the standard representation of surfaces employed within CAD tools [6]. The geometric modeling of continuous developable surfaces remains an unsolved problem, except for specific cases. The techniques developed are based on the optimization of spline-based approximations, the use of dual representations (in which the surface is represented by the envelope of its tangent planes), and isometric mapping between the planar domain and the surfaces [6, 7]. The underline methods establish a connection with the research in computational origami.

Rhinoceros is a CAD tool that offers a variety of resources for computing the flattened state of developable surfaces. According to the official documentation, Rhinos provides the UnrollSrf command to flatten surfaces that are developable. However, the efficacy of this process is contingent upon the complexity of the surface in question. Warped regression edge surfaces are a particular concern that has not been adequately addressed. An exception to this rule may be the developable helicoid, which is a tangential developable surface with constant curvature warped regression edge, that is properly solved (although some exceptions may appear in particular conditions) [8].

Grasshopper, a Rhinoceros plug-in, facilitates the generation of analytical surfaces through the implementation of parametric algorithms. It has been used to implement new approaches to flatten additional developable surfaces, from non-complex straight-edged folded surfaces [9], to algorithms that allow for the determination of the unrolled shape of any developable tangential surface using an approximate discrete strategy [10]. The objective of developable tangential surfaces with warped regression edge was also addressed by Quintial et al. [11].

The objective of this preliminary study is to examine the potential that the CeDG approach offers to model developable surfaces of any type, except for radial surfaces (cylinder or cone), as other studies have previously examined these. Our primary interest lies in addressing the issue of warped regression edges, which may include non-proper (infinite location) zones. Two case studies were selected to guide the analysis.

2 Methods and materials

For the present study, the CeDG has been implemented into the GeoGebra classic version 5 (desktop). GeoGebra is a dynamic geometry software application that can be executed through the GeoGebra.exe executable file or as a web service in the cloud. The software is distributed under the GNU General Public License (v3).

The algorithms designed to generate computational models of developable surfaces are implemented in GeoGebra Scripting code (GGBScripting). The intersection of surfaces and their flattened state is calculated using the extensions of DG methods for CeDG [2, 12]. The two case studies defined to analyze the capabilities of the CeDG approach are presented below.

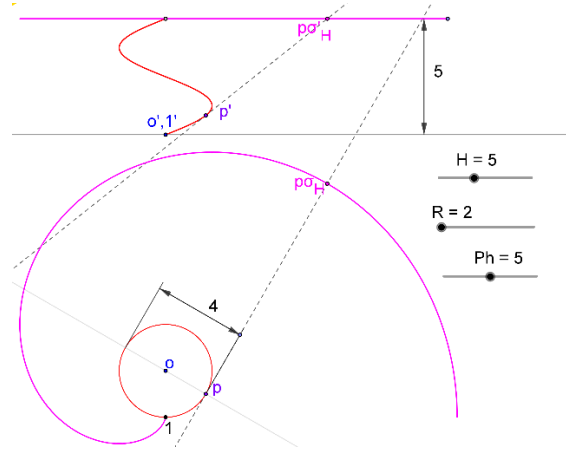


Fig. 1. Developable helicoid with helix pitch, $Ph = 5$ cm, and helix radius, $R = 5$. It shows a generatrix through the point p' - p and the intersection of the helicoid surface with the horizontal plane at $H = 5$ cm.

- i) **Developable helicoid.** This is a tangential surface with generatrix lines tangent to a helix curve, which is also the regression edge of the surface [4].

As shown in Fig. 1, the helix is depicted by its projections, along with the primary parameters that delineate the surface. The figure also shows the intersection of the surface with the horizontal plane at H . This is computed as the locus of the intersection of the generatrix by P (p' - p) with the horizontal plane, $p\sigma H$ - $p'\sigma H$, as Locus ($p\sigma H$, p). This is derived after defining p as a free point on the horizontal projection of the helix, in agreement with the CeDG intersection approach [2].

The helix curve can be modeled in two ways: first, through the locus of a point that fulfills the sum of vertical plus circular movement; and second, by its canonical equation. The latter is employed for the sake of simplicity. Therefore, its vertical projection in Fig. 1 is given as $\text{Curve}(x(o') + R \sin(t), y(o') + Ph \cdot t / (360^\circ), t, 0^\circ, H / Ph * 360^\circ)$, where Curve is a GGB command that provides the curve defined for the implicit variable t . The sliders were created to facilitate the exploration of the surface by changing the values of the main parameters.

- ii) **Convolute with two directrix lines** defined by a planar curve (circle) and a warped curve on the surface of a cylinder.

Fig. 2 shows the planar directrix (horizontal circle with diameter $2 \cdot R_1 = 5.6$ cm) and the warped directrix defined on the cylinder and with circular horizontal projection (diameter $2 \cdot R_2 = 10$ cm), together with the generatrix AB of the convolute. The calculation was performed by first determining the tangent to the warped directrix at point A (a' - a), which is cut by the plane of the first directrix at point Q (q' - q). From there, the tangent to the circle at point B (b' - b) was calculated.

The figure also shows several sliders with the main parameters of the system: R_1 , R_2 (aforementioned) and R_C (diameter of the cylinder), which allow the dynamic exploration of the convolute.

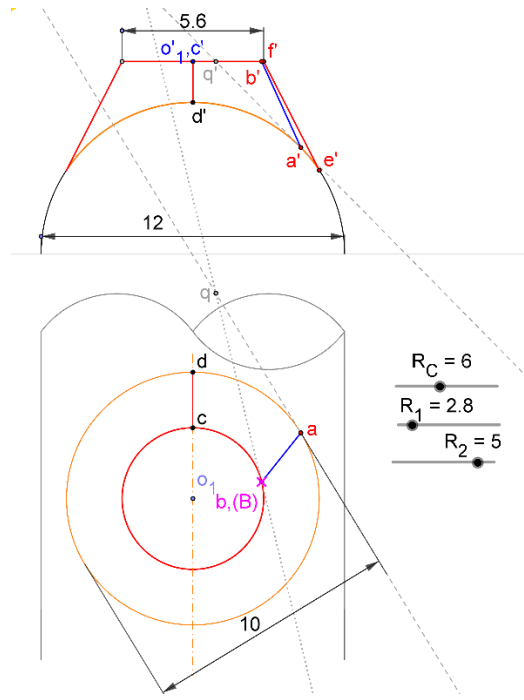


Fig. 2. Convolute with two directrix lines defined by the horizontal circle with center o'_1 - o_1 and radius $R_1 = 2.8$ cm, and the warped curve with circular horizontal projection (radius $R_2 = 5$ cm) on the surface of the revolution cylinder ($R_C = 6$ cm).

The selection of these developable surfaces was driven by the observation that the regression edge of the helicoid displays a constant-radius primary curvature. Thus, it can be utilized as a supporting curve in the flat pattern process. In contrast, the warped regression edge of the convolute has a variable curvature and includes extension to infinite (non-proper definition).

3 Theoretical analysis under descriptive geometry

The modeling of a general developable surface under DG procedures is based on the movement law that a plane tangent to each generatrix executes. This law can be implemented in a straightforward manner in GeoGebra, thereby providing a method to generate surface projections in a canonical position in CeDG. Figures 5, 6, and 8 illustrate this technique, presenting a discrete number of generatrices for the developable helicoid and the convolute defined.

Given any curve C in the developable surface and two points A and B on it, the process of flattening C utilizes the invariance of the distance between A and B and the invariance of the angle between the tangent to C by A (or B) and the generatrix through A (or B) [4] associated with the developable feature. The strategy is to find a curve S in the surface with a known flattened state (support curve), and to place the surface's generatrices in the flattened state through their intersection with that curve.

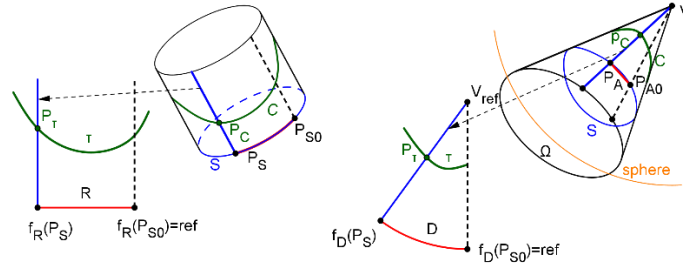


Fig. 3. Support curves, S , in a cylinder (left) and cone (right) with their flattened transform (R in cylinder and D in cone). An arbitrary generatrix is transformed into flat space through the incidence point with S (P_S). Other curve C is flattened through the distance from its point in the generatrix (P_C) to P_S .

This strategy is the gold standard in the case of cylinders and cones, as shown in Fig. 3. The supporting curve is a right section in the cylinder or the intersection with a sphere [4] in the cone¹. However, there is not a support curve for a general developable surface.

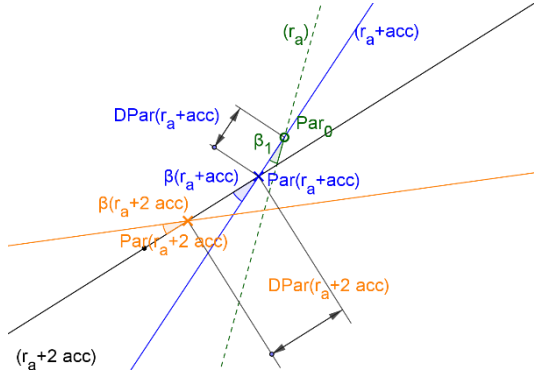


Fig. 4. Generatrices by r_a , r_a+acc , and $r_a+2 acc$, where $acc \ll 1$ (see text).

Given that the regression edge of a developable surface—the curve generated by the tangency to all the surface generatrices—is the unique curve in the surface that keeps the flexion curvature when it is flattened, it is possible to develop an algorithm that computes its flattening state. In turn, any other curve on the developable surface can be flattened by using the regression edge as the support curve.

Fig. 4 presents a strategy for transforming the regression edge to the flat space through a

discrete number of points in it. Any point on a directrix curve of the developable surface is defined as $P(r_a)$, where r_a is a real variable in $[0,1)$. The intersection of generatrix by

¹ The sphere method for the flattening of cones is avoided in the classical DG treatises, which were oriented to the paper version, in favor of the approximated pyramid method.

$P(r_a)$ to generatrix by $P(r_a+acc)$, where $acc \ll 1$ (e.g., $acc = 10^{-3}$), gives the point $Par0$ in the regression edge. This is regarded as a reference point in flat space.

Starting with reference point $Par0$, the next point in the regression edge, $Par(r_a+acc)$, is obtained by measuring the distance to the previous $DPar(r_a+acc)$ and the angle between the generatrices, $\beta(r_a+acc)$. This process is repeated to ensure that the necessary portion of the regression edge is covered. It can be demonstrated that the set of points $Par(r_a + i \cdot acc)$ for $i = 1, 2, \dots, n$ converges to the true regression edge when acc tends to 0 and n tends to infinity, provided $acc \cdot n = \text{length of regression edge}$, for a finite regression edge.

In cases where the regression edge is not finite, it may be possible to extend this strategy by considering the parallelism between the generatrices and the regression edge asymptote.

4 Developable helicoid modeling

Fig. 5 (left) shows the projections of one of the leaves of the developable helicoid of Fig. 1, which is limited by the horizontal plane H . Generatrices segments limited to the leaf and defined by points from the helix (regression edge) spaced 5° in the horizontal plane are presented.

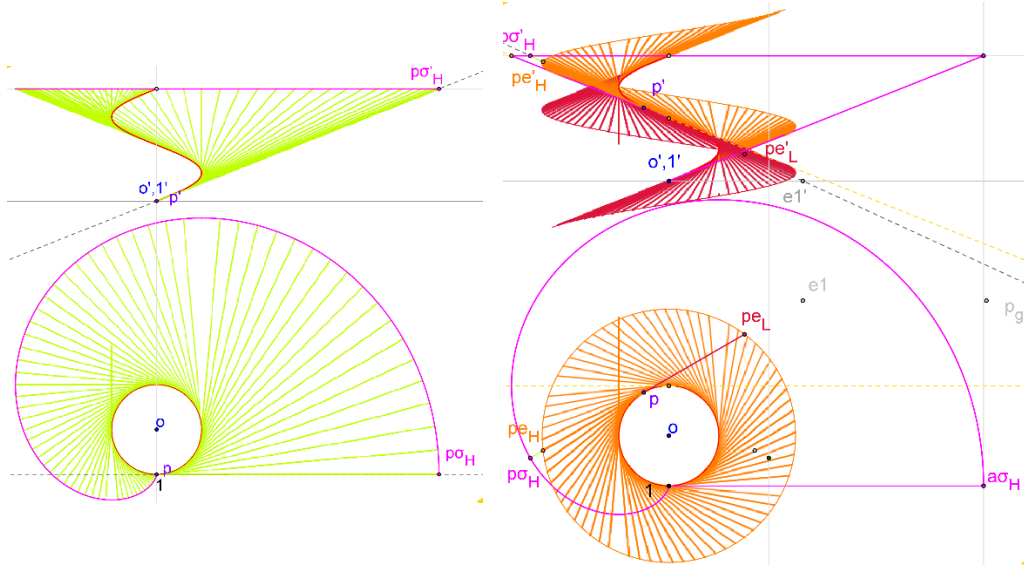


Fig. 5. Projections of the developable helicoid of Fig. 1, limited by horizontal planes (left) and with fixed generatrices length (right).

A more practical case is presented in Fig. 5 (right), where the generatrices are limited by a fixed length. The vertical projection displays both leaves with different colors, while the horizontal projection shows only the upper leaf. This surface is used in the design of worm gears and several types of helicoid bearings, for example.

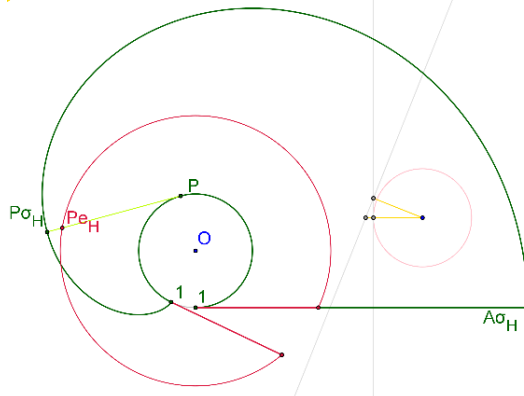


Fig. 6. Flattened helicoid (one leaf) of helicoids of Fig. 5. The circle with center O is the flattened state of helix, whose radius is computed through the reduced step triangle (diagram is sketched) [4]. Generatrices are transformed through their tangency to the circle.

Due to the constant curvature of the helix, its flat transformation can be obtained using this one as the support curve. Fig. 6 shows the flattened state of the developable helicoid limited by horizontal planes for the leaf shown in Fig. 5 - left (Fig. 6 – green), and the flattened state of the developable helicoid with fixed generatrix length for the higher leaf in Fig. 5 - left (Fig. 6 - orange).

5 Warped directrix convolute modeling

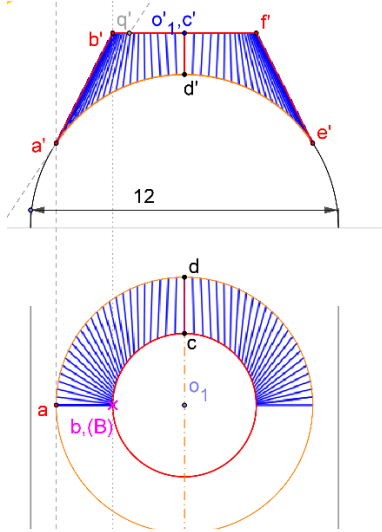


Fig. 7. Orthogonal projections of the convolute surface of Fig. 2.

Fig. 7 shows the projections of the convolute of Fig. 2 between the two directrix curves, including a discrete number of generatrix segments for the anterior half surface (taking advantage of the convolute symmetry).

Generatrices are modeled using the procedure outlined in the methods and materials section, with $a = \text{Point}(h, r_a)$, where h is the horizontal projection of the warped directrix, and r_a the mapping variable in $[0,1)$ (see theoretical analysis above). Fig. 7 presents 50 generatrices computed from $r_a = 0$ to 0.5 (increment 0.01). A visual exploration of their behavior reveals that generatrices near the frontal plane by o'_1-o_1 intersect at points close to the directrix curves, whereas the intersection points extend away for those near the DC generatrix.

The regression edge, computed as the locus of the intersections of the generatrices through $\text{Point}(h, r_a)$ and $\text{Point}(h, r_a + \text{acc})$, when r_a falls within the interval $[0,1)$ and $\text{acc} = 10^{-3}$ gives the curves defined by the projections in Fig. 8.

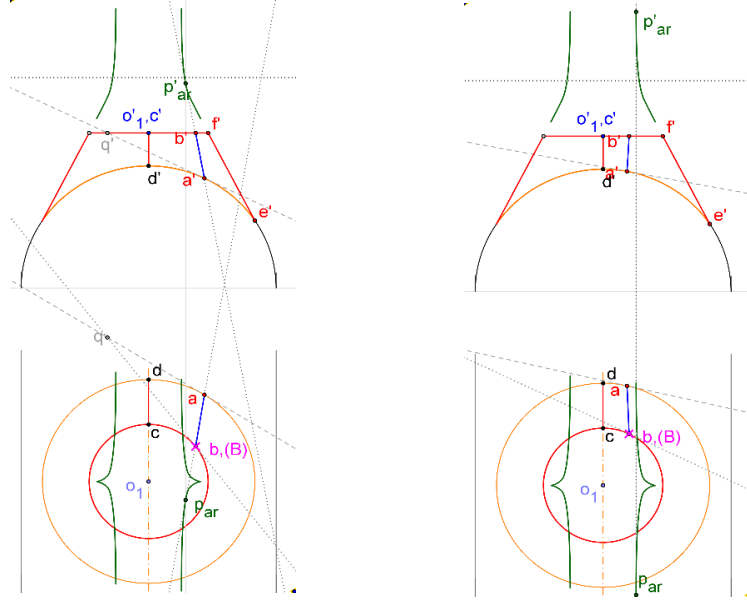


Fig. 8. Regression edge (green) of convolute of Fig. 2, by its projections for r_a in $R_p = [0,0.2] + [0.3,0.7] + [0.8,1)$, and $acc = 10^{-3}$, together with the generatrix AB, for A associated with $r_a = 0.162$ (left) and $r_a = 0.214$ (right).

According to the findings of the exploration, the regression edge does not converge in the proximity to CD generatrix. However, it is possible to compute the proper regression edge in the R_p zone. Fig. 8 shows a generatrix in the convergence zone (left) and in the non-convergence zone (right). As seen, the associated regression edge point p'_{ar} moves away from directrix curves when r_a increases from 0.162 to 0.214 (it has been restricted to the limit value $r_a = 0.2$ due to the divergence).

We have implemented the previous outlined design strategy to achieve the flattened state of the surface. According to the symmetry of convolute, it is sufficient to calculate it from FE to CD generatrices (see Fig. 9), where r_a falls within $[0,0.25]$. We have folded down the plane defined by two adjacent generatrices to find the flexion angle between them and rotated the generatrix to measure the distance between regression edge points. These standard DG procedures are presented in Fig. 9. Once these ones were applied to a generic generatrix parameterized by r_a , we built two GeoGebra tools. A more detailed description of the technique exceeds the scope of this chapter.

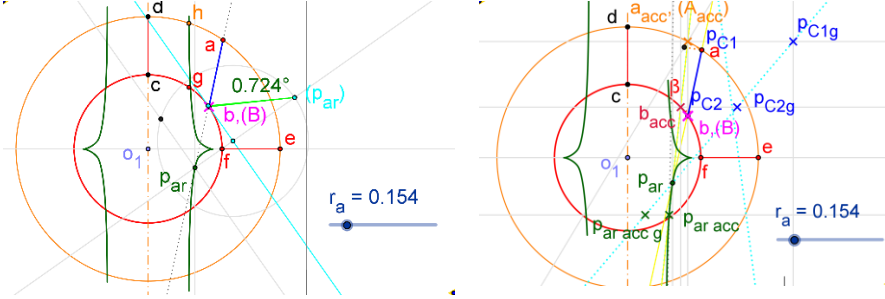


Fig. 9. Zoom to constructive DG methods to obtain the angle between adjacent generatrices (left) and distance between adjacent regression edge points (right).

It is worth noting that due to the inherent challenges of working with very close generatrices, the methods were applied with $\text{acc} = 0.02$ instead of $\text{acc} = 10^{-3}$. Once the tools have been created, simply use the desired value of $\text{acc} = 10^{-3}$, or other minor value if greater accuracy is required, as argument to the tools. Fig. 9 left presents the system for $\text{acc} = 10^{-3}$, whereas Fig. 9 - right does with $\text{acc} = 0.02$. As expected, the quality of the regression edge solution for $\text{acc} = 0.02$ is inadequate and the curve is not symmetrical.

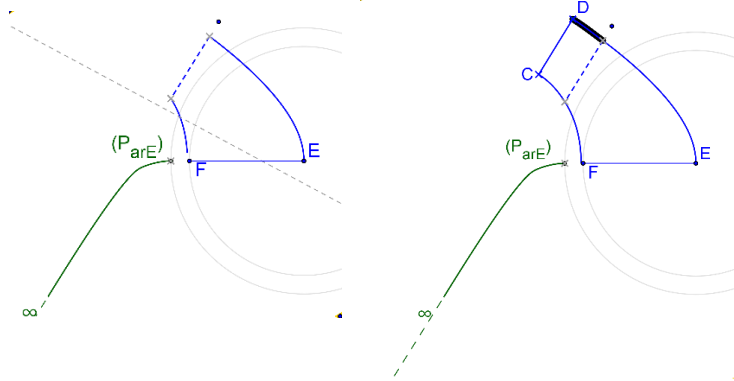


Fig. 10. Flattened surface ($\text{acc} = 10^{-3}$) for the proper zone of regression edge, (r_a in $[0, 0.2)$, left) and full flattened surface (r_a in $[0, 0.2)$, right). See text.

Fig. 10 – left shows the flat transformed of the convolute (a quarter) using the proper zone of regression edge as support curve.

While the regression edge is not accessible in the non-proper zone, we are able to ascertain its asymptote. As generatrices are parallel to it, the flat transformation of the directrix curves were calculated through the invariance of the angle between the convolute generatrix and their tangents at each point of the directrix curve. This is shown in Fig. 11, where the planes defined by the generatrix and tangent to each curve are folded down to horizontal planes, to solve the angle in the plane directrix, β , and the angle in the warped directrix, γ .

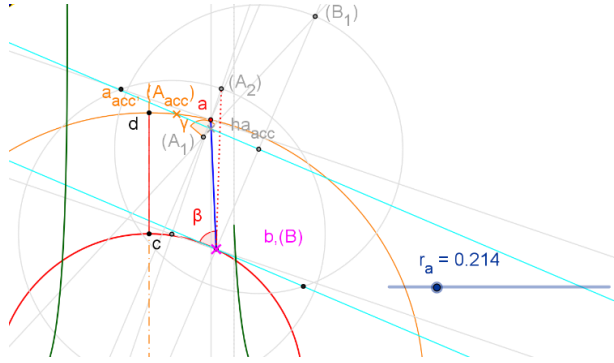


Fig. 11. Zoom to the constructive DG method to obtain length of tangents segments and their angle with the convolute generatrix ($\text{acc} = 0.02$, see text).

The lengths of the small segments that define the tangents to each curve are calculated through their true size in the folded-down plane. The notation (A_1) is followed by the folded down of A point as pertaining to the plane tangent to the warped directrix, whereas (A_2) is used for the folded down of A as pertaining to the plane tangent to the plane directrix. The condition of tangency of a

segment to each curve is associated with its small length, which demonstrates the connection to differential geometry. The DG method is employed again with $\text{acc} = 0.02$ to streamline the graphical manipulation. The full flattened surface is shown in Fig. 10 – right. The flat transformed of the warped directrix in the non-proper regression edge zone is shown as a thickened curve to illustrate that the curve is obtained through spline interpolation into discrete points.

6 Discussion and conclusions

The parametric 3D model of the helicoid with its flat pattern has been successfully resolved using standard CeDG techniques, yielding an exact solution. A novel algorithm has been developed for the convolute surface to automate the flat pattern of the regression edge through discretization. This curve provides support for the generatrices and, as a result, for the flat pattern. The convolute's flat pattern, unlike the helicoid, is discrete and non-exact. However, its precision can be increased as needed through the parameter acc . Nonetheless, the discretization technique prevents the dynamic parametrization that characterizes CeDG. Therefore, the user must click on a button (see Fig. 12) to recalculate the flat pattern.

This algorithm is subject to one additional limitation: its convergence region, R_p , is dependent on the model's parameters. As demonstrated in Fig. 12, the flat pattern of the convolute is shown for the same model when R_1 decreases to 2.3 cm. As evident, the CD generator is not parallel to the regression edge asymptote (deviation of 3.5°).

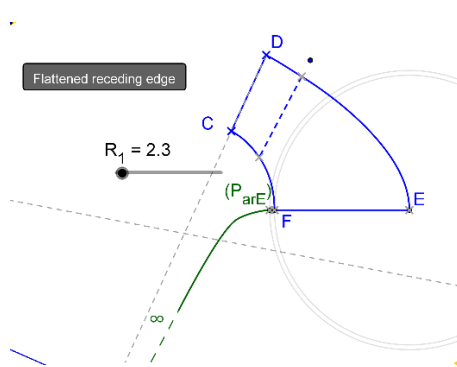


Fig. 12. Flat pattern of the convolute (one quarter) when R_1 is decreased to 2.3 cm (see text).

As illustrated in Fig. 13, this deviation can be rectified by exploring the solution of the regression edge.

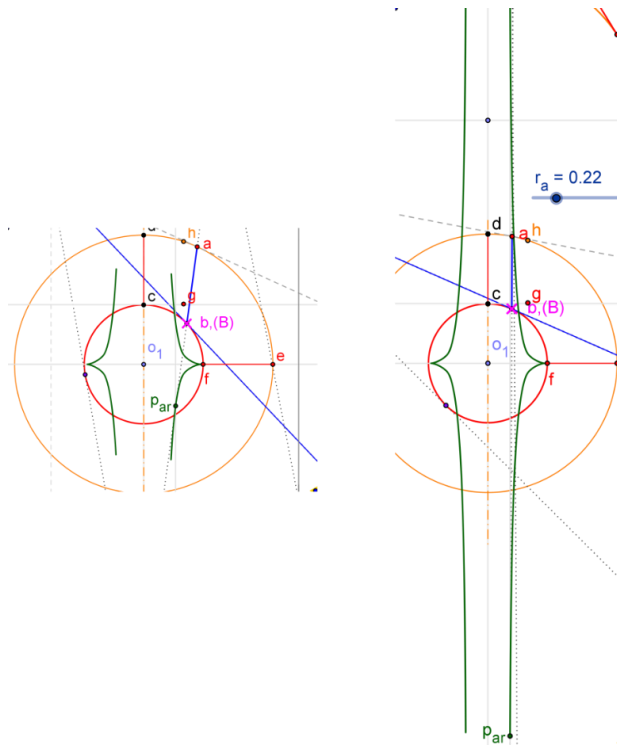


Fig. 13. Horizontal projection of the regression edge of convolute of Fig. 2 ($\text{acc} = 10^{-3}$), after changing R_1 from 2.8 to 2.3 cm, for $R_p = [0, 0.2] + [0.3, 0.7] + [0.8, 1)$ (left) and $R_{px} = [0, 0.22] + [0.28, 0.72] + [0.78, 1)$ (right).

This figure shows the extension of the proper zone of the regression edge when R_p extends from the initial definition, R_p , in Fig. 8 to the region R_{px} . The horizontal projection of the regression point Par ($r_a = 0.22$) can be found (Fig. 13 – right). A minor increase from $r_a = 0.22$ will result in an abrupt departure of Par .

In summary, this preliminary study has demonstrated the effectiveness of CeDG in computing flat patterns of general developable ruled surfaces with warped, complex regression edges. Further research is necessary to automate the computation of R_p , manage composite developable surfaces (that require global developability due to overlapping [6]), and to compare the accuracy and robustness of the final approach to other CAD tools, such as Rhinoceros, with the aim of applying it to the real-world design.

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